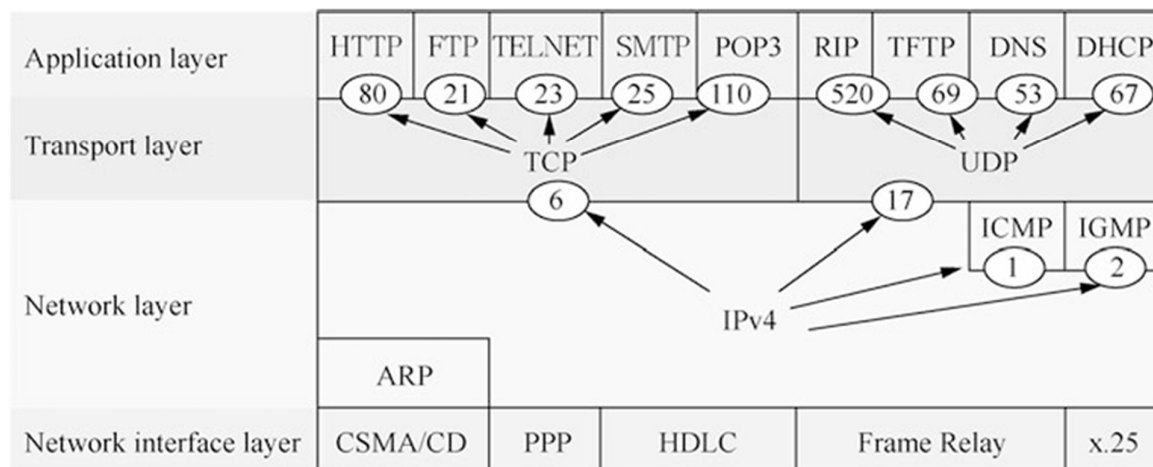


Fig. 2.24 Well-known port numbers

HTTP	FTP	SMTP	POP3	TELNET	RDP	DNS	RIP	TFTP	DHCP
(80)	(21)	(25)	(110)	(23)	(3389)	(53)	(520)	(69)	(67)
TCP						UDP			

**Fig. 2.32** TCP/IPv4 protocol stack

2.4 Network Layer Protocols

55

Protocol name	ICMP	IGMP	IP	TCP	IGP	UDP	IPv6	ESP	OSPF
Protocol field value	1	2	4	6	9	17	41	50	89

Fig. 2.35 Commonly used protocols and corresponding protocol field values

Class A	0NNNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	0.0.0.0–127.255.255.255	Assigned to hosts
Class B	10NNNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	128.0.0.0–191.255.255.255	
Class C	110NNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	192.0.0.0–223.255.255.255	
Class D	1110NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	224.0.0.0–239.255.255.255	Used for multicast
Class E	1111NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	240.0.0.0–255.255.255.255	Used for research

- Default subnet masks of classes A, B, and C
 - Class A: 8 bits, 0.0.0.0–127.255.255.255/8
 - Class B: 16 bits, 128.0.0.0–191.255.255.255/16
 - Class C: 24 bits, 192.0.0.0–223.255.255.255/24

Network part
Host part

Private IP address: In practice, some networks do not need to connect to the Internet. For example, on a network of a lab in a college, IP addresses of devices need to avoid conflicting with each other only within the same network. In the IP address space, **some IP addresses of class A, B, and C addresses are reserved** for the preceding situations. These IP addresses are called **private IP addresses**.

- Class A: 10.0.0.0–10.255.255.255
- Class B: 172.16.0.0–172.31.255.255
- Class C: 192.168.0.0–192.168.255.255

Special IP Address	Address Scope	Function
Limited broadcast address	255.255.255.255	It can be used as a destination address and traffic destined for it is sent to all hosts on the network segment to which the address belongs. (Its usage is restricted by a gateway).
Any IP address	0.0.0.0	It is an address of any network. Addresses in this block refer to source hosts on "this" network.
Loopback address	127.0.0.0/8	It is used to test the software system of a test device.
Link-local address	169.254.0.0/24	If a host fails to automatically obtain an IP address, the host can use an IP address in this address block for temporary communication.